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RESEARCH ARTICLE

Different faces of cigarette butts, the most abundant beach litter worldwide

Maria Christina B. Araújo[1] · Monica F. Costa[2] · Jacqueline S. Silva-Cavalcanti[3] · Armando C. Duarte[4] · Vanessa Reis[4] · Teresa A. Rocha-Santos[4] · João Pinto da Costa[4] · Violeta Girão[4]

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Abstract

Cigarette butts (CBs) are non-biodegradable residues of synthetic origin, prevalent on beaches all over the world. The study evaluates discarded CBs on an intensely used urban beach, determining variations in physical and chemical characteristics. CBs collected were observed, classified, and visually separated according to a proposed scale of four levels of degradation to test the potential match between physical and chemical decay. CBs (un-smoked, smoked, and discarded) were used to determine the average length (cm) and mass (g) in order to observe changes in these parameters among the levels. Cigarette butts experience consecutive mass loss during environmental exposure. Scanning electron microscopy (SEM) images were obtained to assess physical changes in fibers due to smoking. FTIR-ATR was used to assess CBs new (un-smoked), smoked, and discarded samples in relation to cellulose acetate decay. The FTIR-ATR spectroscopy of the most visually degraded cigarette butts indicated modifications in the spectra when compared to un-smoked cigarettes.

Keywords Marine pollution · Beach litter · Microplastics · Monitoring · Cellulose acetate · Environmental degradation

Introduction

Litter accumulation in coastal areas has been reported and highlighted as a growing threat by researchers in several countries. Urban beaches with high recreational use are full of plastic items and cigarette butts (CBs) (Ribeiro et al. 2021; Araújo and Costa 2021; Araújo and Costa 2019a; Maziane et al. 2018; Silva et al. 2008a; UNEP, 2009; Hidalgo-Ruz et al. 2018; Becherucci et al. 2017).

It is estimated that 5.5 trillion cigarettes are consumed globally each year, and that 4.95 trillion CBs are inappropriately discarded somewhere in the environment worldwide (http://cigwaste.org; Novotny and Slaughter 2014), including beaches. Smokers often discard CBs in public places inappropriately (Novotny and Slaughter 2014; Patel et al. 2013; Schultz et al. 2011; Smith and Novotny 2011; Rath et al. 2012).

Quantitative data generated annually by Ocean Conservancy during international coastal cleansing events have demonstrated the magnitude of the problem. In the 2018 campaign, for example, volunteers worked on beaches in more than one-hundred countries, collecting a total of 5,716,331 CBs, an amount that placed this type of waste at the top of the list of most abundant items of litter (Ocean Conservancy, 2019). It is worth mentioning that this fact has been occurring systematically since the campaign went global in 1991. Numerous factors are involved in a convergent way to determine the high presence of this type of waste on beaches. The habit of smoking cigarettes (widespread and tolerated worldwide) and the irregular disposal of CBs in public spaces undoubtedly have a relevant weight in the number of CBs that are detected in coastal environments (Araújo and Costa 2021, 2019a).

CBs are predominantly non-biodegradable synthetic residues, prevalent in coastal environments in various parts of

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✉ Maria Christina B. Araújo
mcbaraujo@yahoo.com.br

[1] Departamento de Oceanografia e Limnologia, Universidade Federal Do Rio Grande Do Norte, Natal, Brazil

[2] Departamento de Oceanografia, Universidade Federal de Pernambuco, Recife, Brazil

[3] Departamento de Biologia, Universidade Federal Rural de Pernambuco, Recife, Brazil

[4] Departamento de Química, Universidade de Aveiro, Aveiro, Portugal

the world (Shen et al. 2021). They have a high potential for impacting on a wide range of socioeconomic and environmental aspects (Araújo and Costa 2019b). CBs can be reported as an isolated waste category, mainly due to their specific composition and use (Bravo et al. 2009; Smith et al. 2014; Blickley et al. 2016; Becherucci et al. 2017; Hidalgo-Ruz et al. 2018), but also to call attention to the available amounts found on beaches. Currently, European Chemicals Agency (ECHA) is considering including CBs as microplastics (MPs) in the directive on single-use plastics. These residues are composed of a fibrous material made of cellulose acetate (polymer obtained from cellulose plus the addition of acetic anhydride and acetic acid) coated with paper and tobacco. Most of these filters are made of more than 15,000 strands (Belzagui et al. 2021). Their size usually does not exceed three centimeters, and they have a light color, both internally and externally. Some factors favor their accumulation and permanence in coastal environments. Their size and color facilitate remaining unnoticed mixed in the sands of the beaches, making it difficult to be removed by manual or mechanical processes (Ariza et al. 2008). In addition, cellulose acetate is characterized by extremely slow degradation, even under severe environmental conditions (Puls et al 2011; Haynes et al. 1991; Belzagui et al. 2021). It is worth mentioning that this type of waste has a very low density, a fact that allows CBs to be transported by water or wind beyond the places where they were discarded (Armitage and Rooseboom 2000; Williams and Simmons 1999). CBs also are a new source of microplastic fibers, as they can be gradually decomposed into small plastic fibers. The discarded cigarette butts can increase the accumulation of microplastic fibers in the environment (Belzagui et al. 2021; Shen et al. 2021).

More than 5,000 compounds (from tobacco and added substances) can be found in a cigarette, of which at least 150 are recognized as highly toxic due to their carcinogenic and mutagenic potential (Shen et al. 2021; Moriwaki et al. 2009; Hoffmann and Hoffmann 1997, 1998; Slaughter et al. 2011). The compounds with the greatest toxic potential are concentrated mainly in the remains of the tobacco and in the filters (Slaughter et al. 2011; Barnes 2011). The substances that make cigarettes potentially dangerous for smokers (active and passive) and have the potential to cause environmental damage are incorporated throughout the product life cycle, from tobacco planting to the completion of the cigarettes for sale (Hoffmann and Hoffmann 1998; Sheets 1991; Araújo and Costa 2019b).

Two chemical processes are mainly responsible for the impacts of CBs on natural environments, namely, (a) leaching of compounds present in CBs by rainwater and (b) the transfer of these compounds to water bodies through urban runoff (Slaughter et al. 2011; Araújo and Costa 2019a, b). The main compounds include nicotine, polycyclic aromatic hydrocarbons, and metals (Buerge et al. 2008; Moriwaki et al. 2009; Valcarcel et al. 2011; Green et al. 2014; Dobaradaran et al. 2017). In the environment, these substances come into contact with the biota, which can be largely affected by the toxicity of the compounds (Savino and Tanabe 1989; Micevska et al. 2006; Rawls et al. 2011; Di Giacomo et al. 2015; Booth et al. 2015; Wright et al. 2015; Slaughter et al. 2011; Lee and Lee 2015), or by ingestion (Macedo et al. 2011). An interesting interaction between cigarette butt waste and wildlife is the incorporation of littered butts into the building of nests by passerine bird species, but the effect (if any) of this behavior remains unclear; probably nicotine may be acting as arthropod repellents (Suárez-Rodríguez et al. 2013; 2017).

Probably, even in CBs with no smoke remains, toxic substances are still present in the fibers of the filters, especially in those recently discarded, with the possibility of being spread into the environment (Araújo and Costa 2019a, b). Therefore, it is important to assess the degradation processes of this type of waste. The information obtained can be used to determine the potential for toxicity over time, and substances that last longer. The possibility of CBs causing the same effect than microplastics cannot yet be ruled out, i.e., adsorbing organic pollutants available in the environment.

The perceived degradation of CBs deposited on the beach sand observed with the naked eye might not necessarily mean that they are being chemically modified. The present study assessed new (un-smoked), smoked, and discarded CBs, to determine if variations in their physical (visual aspect, length, and mass) aspect were related to detectable chemical change (decay of cellulose acetate).

Materials and methods

Sample collections (discarded CBs) were performed in February 2018 in the extreme north of Boa Viagem Beach (Fig. 1), located in the northeast of Brazil, in the city of Recife (Pernambuco), which has a population of over 1,500,000 inhabitants. Boa Viagem represents the most accessible coastal space for the local population, as well as for national and foreign visitors, being usually intensely frequented, especially in the summer months. A total of 3,020 users have already been registered in a 100 m beach segment in December, for instance (Silva et al. 2008b). According to a beach classification system used in Brazil, which evaluates landscape typology and settlement intensity (Araújo and Costa 2020), Boa Viagem is classified as "exposed seaside with consolidated urbanization," falling into the category of "very developed beach." The beach is cleaned every day. The cleaning process occurs both manually (from 7a.m. to 5p.m.) and with a mechanized sieve (4a.m.), with an average mesh size of 3 cm.

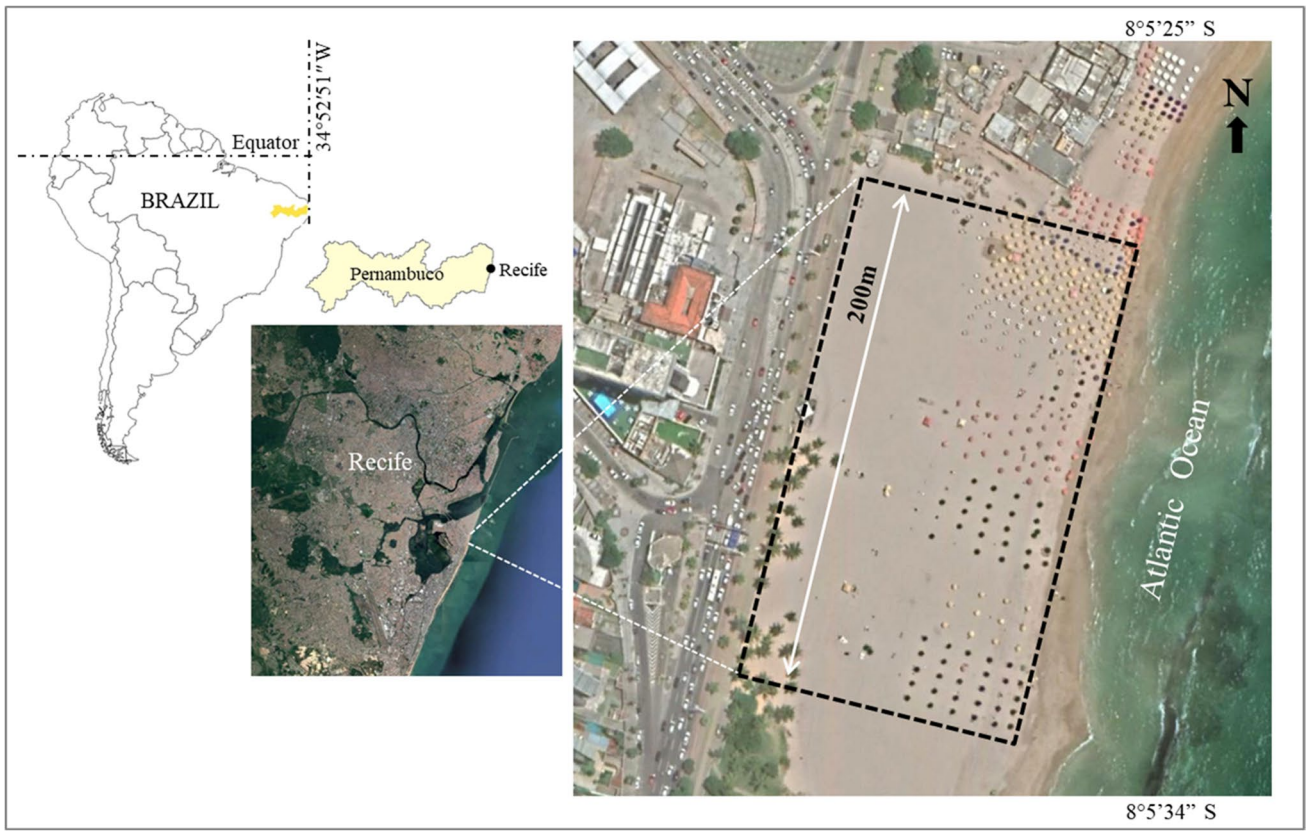


Fig. 1 Boa Viagem Beach at Recife, Brazil. Location of the intensely used beach area where CBs were collected for the present study

The total study area had 20,000m^2. Each of the ten transects had 100m^2. Then, the sampled area was 1,000m^2. Transects were parallel to the coastline, extending from the boardwalk to the waterline. All CBs visible on the surface of the sand within the transects were collected using stainless-steel tweezers, stored in aluminum containers with the respective labels, and taken to the laboratory for analysis. At the lab, samples were oven-dried them at 40 °C overnight and cleaned with a soft brush to remove any remaining adhered sand.

Un-smoked cigarettes were purchased at popular vendors. Packs were opened, and cigarettes ($N = 20$) were smoked by a single volunteer (smoked).

Degradation scale

The CBs collected in transects were observed and visually classified into degradation levels. After sorting and separating the CBs into the established degradation levels, CBs (smoked and discarded) were used to determine the length (cm) and mass (g) averages in order to observe possible differences in these parameters among the levels. Un-smoked filters were measured only for reference of the starting length and mass of the cigarette filters used in the experiment. Chemical analysis followed to instrumentally test the degradation of each level of degradation separately.

Chemical analysis

The types of CBs chemically assessed in the present study were new (un-smoked), smoked, and discarded CBs (these last divided into four visually determined degradation levels I to IV).

Scanning electron microscopy (SEM) images were obtained with a field emission gun (FEG)—SEM Hitachi S4100 microscope (Japan), operated at 25 kV. Samples were prepared by deposition directly onto the carbon tape and then coated by carbon evaporation (Emitech K950X, France).

In a FTIR-ATR Spectrum 400 (Perkin-Elmer; scanning: 16; resolution: 4 cm^{-1}; band: 4000–700 cm^{-1}), reference (Supplementary Electronic Material) and acquired spectra for cellulose acetate were used for comparisons with the un-smoked, smoked, and discarded CBs.

Statistical analyses were conducted using MATLAB®. The principal components analysis (with 95% confidence level) was used to compare the average spectra of the three

types of samples, including un-smoked, smoked, and discarded in the four degradation levels (I to IV).

Results

Degradation scale

A total of 241 CBs were collected from the beach. The observation made it possible to determine four levels of degradation, namely, (I) with tobacco residues and preserved filter and paper coating; (II) without tobacco residues, preserved filter and paper coating, and manufacturer's identification; (III) no tobacco remains and filter with paper coating, but considerably worn and discolored, without manufacturer's identification; and (IV) only the fibers of the flat and compacted filter, without paper coating (Fig. 2a). The greatest variation related to the visual characteristics of the CBs was found among those at level IV (Fig. 2b). Of the total of CBs collected, those classified at levels III and IV were predominant, with 31.49 and 47.51%, respectively. Levels I and II appeared in smaller quantities, with 11.60 and 9.39%, respectively.

Samples of un-smoked clean filters ($N=20$) of the cigarettes used in the experiment, not considering any tobacco and paper, had 2.27 ± 0.17 cm in length and 0.16 ± 0.01 g in mass. The determination of average length and mass of the CBs classified by levels indicated that there was a reduction in both parameters, from smoked and degradation level I toward degradation level IV (Fig. 3).

Chemical analysis

Physical fragmentation could be observed in the SEM images when comparing un-smoked and smoked filters. Discarded CBs also showed physical differences from un-smoked and smoked ones when observed through SEM imaging (Fig. 4).

When comparing the FTIR-ATR spectra of the three types of samples, the degradation level IV showed a separation from the other groups (Fig. 5). In this analysis, PC1 explained 68.71% of the variation, while PC2 explained 16.30% (85.01%). Furthermore, the degradation level III was the second most different from the reference un-smoked CBs (Fig. 6). In this case, PC1 explained 42.58% of the variation and PC2 35.58% (78.16%). Both analyses resulted significant with 95% confidence interval.

Discussion

It is common to see CBs in the sand in this area; however, the amount collected (241) in the transects was slightly less than expected. As only the CBs visible on the surface of the sand were collected, the amount buried should have been much greater. There should probably be a large amount of buried or camouflaged butts, due to their size, color, and compaction process after long periods of accumulation and exposure. These factors usually hinder their removal by manual or mechanical actions, increasing the probability of remaining in the environment (Ariza et al. 2008; Araújo and Costa 2021). This beach area has a mechanized system for removing waste, but the 3-cm mesh retains the least amount

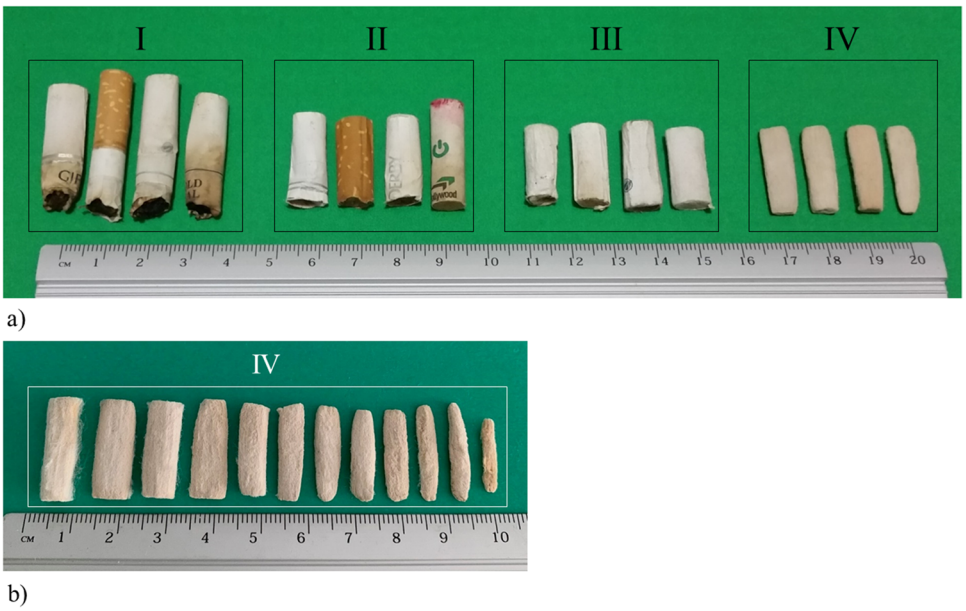


Fig. 2 Different visual characteristics observed in cigarette butts classified them into a scale of four degradation levels I to IV. (**a**) Levels of degradation observed in cigarette butts collected on the beach and characteristic examples of each one; (**b**) examples of the variation within level IV

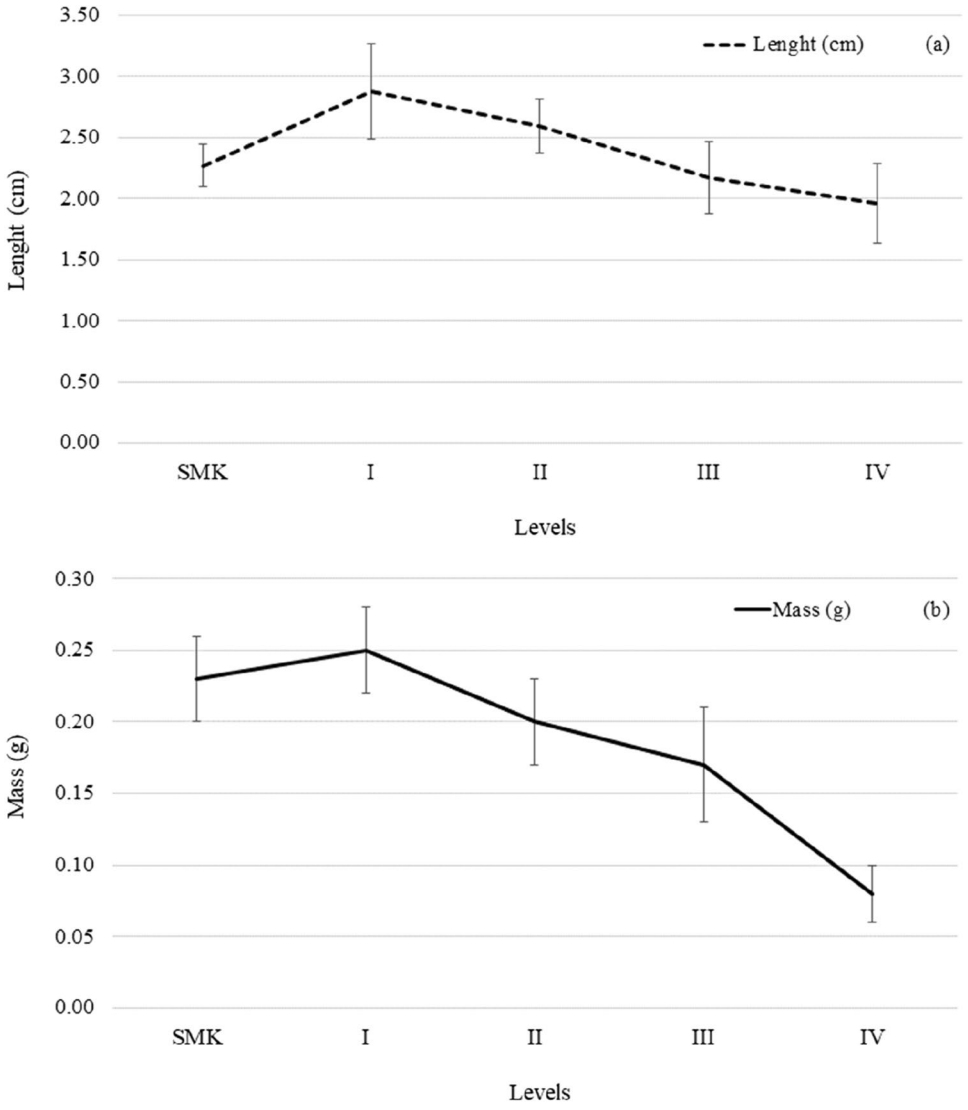


Fig. 3 Average ± stdev. (**a**) length (cm) and (**b**) mass (g) of smoked and discarded CBs per level. SMK $N = 20$; I to IV $N = 20$. Detailed information in the supplementary material provided

of sand possible, which, unfortunately, allows the loss of items that pass through and remain buried in the sand.

In addition, the collection for the present study was performed in the early morning to avoid the hours with the highest concentration of users, who are considered to be direct sources of CBs to the beach. This fact may have been responsible for the reduction of CBs in number.

The four different levels of degradation observed for the collected CBs suggest different periods of exposure to environmental conditions. The present work considered samples available on a tropical beach, without knowing for how long they have been exposed to those environmental conditions. However, comparing the physical state (visual) and chemical degradation (images and FTIR) detected to the nearest condition found in the literature (Bonanomi et al. 2020), it is possible to roughly estimate that level IV CBs might have been on the beach for approximately 5 years. According to these authors, CB mass loss followed a three-step decomposition pattern. A rapid mass loss consisting of ~ 15 and 20% of initial mass was verified in the first month of incubation (Bonanomi et al. 2020). This level of mass loss corresponds roughly to the loss observed between our levels I (0%), II (15%), and III (26%). In the following two years, however, decomposition was very slow reaching a total mass loss of ~ 30 and 35%. Starting from the third year of incubation, a treatment dependent mass loss clearly emerged, with significant differences compared to the first two decomposition phases and consistent until the fifth year of observation (Bonanomi et al. 2020). Our results corroborate with these findings since CBs in level IV in our study had a mass loss of 65%.

Many CBs are commonly discarded with remains of tobacco, a material that is quickly lost, probably in a short time. Tobacco remnants separated from the CBs become another toxic component that may spread through the environment causing additional impacts. Except for the presence

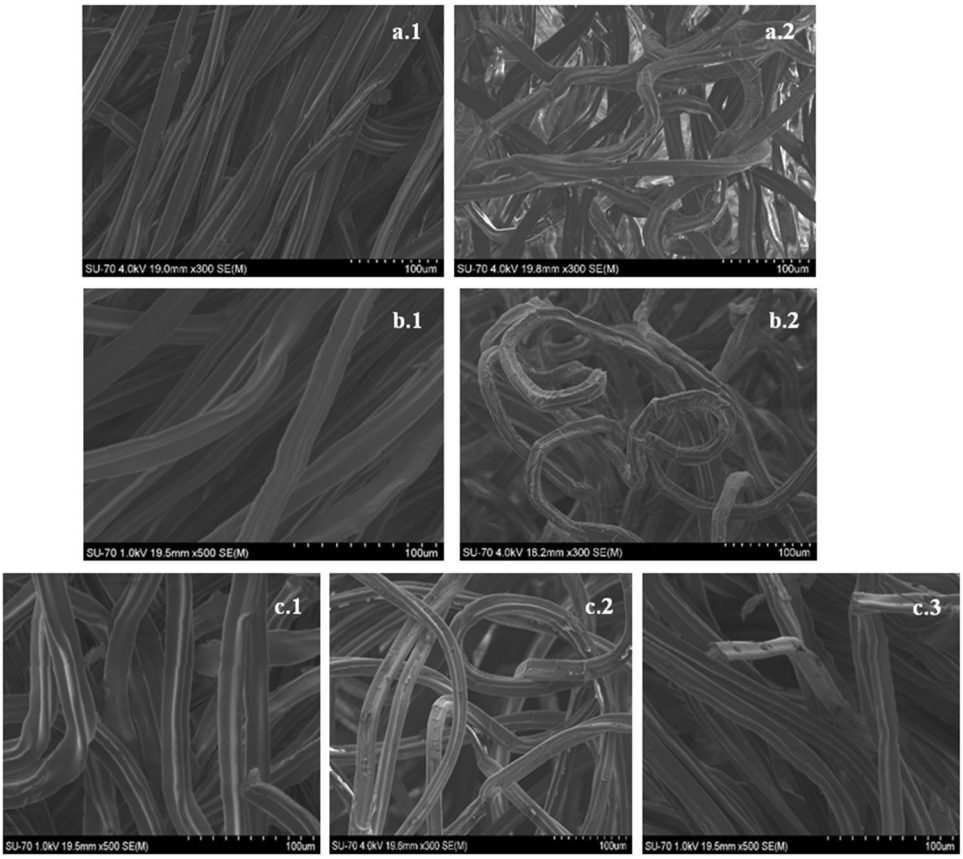


Fig. 4 Examples of SEM micrographs series. (**a**) 1. un-smoked and 2. smoked; (**b**) 1. un-smoked and 2. smoked; (**c**) 1. un-smoked, 2. smoked, and 3. discarded

of tobacco, CBs at levels I and II had some similarities, such as practically intact paper coating, and the possibility of identifying the manufacturers' brands. However, as paper is the most fragile part of CBs, it is quickly influenced by the weather and eliminated.

Throughout some degradation process, CBs move through degradation levels I to IV, and the modification is mainly related to the compaction of the fibers, generating volume, length, and mass losses. The predominance of CBs at level IV (~47%) demonstrated this fact. Progressive reduction, both in mass and length of CBs, can be probably related to the weathering processes generated in the disposal environment. After the loss of paper coating, there was a clear change in the filters, with evidence of fiber release and gradual length and mass reduction (Fig. 2b). The apparent process of fiber release can be probably due to the action of friction with the beach sand. CBs dry out and become rigid and compacted. According to Puls et al. (2011), the total disintegration of conventional cigarettes is hampered by the high compaction of fibers and the addition of plasticisers. It is worth mentioning that, as a result of this process, the higher the level of CBs degradation, the greater the possibility of remaining unnoticed mixed in the beach sand.

Bonanomi et al. (2015) assessed the degradation process of CBs in the environment. The results confirmed the low rate of degradation, given that, on average, they lost only 37.8% of their initial mass after two years of decomposition. These authors observed that, after the rapid deterioration of the small labile fraction, corresponding to the outer paper layer, the degradation proceeds very slowly, regardless of the environmental conditions present during the process.

SEM images illustrate the typical structural features of CBs having triangular cross section of fibers. Cigarette filters consist of thin and long continuous fibers, entangled to filter vapors and particulate matter generated during cigarette combustion. In addition, over time, the design of the fibrous morphology has been optimized. Triangular cross-section fibers are being used to minimize health issues. The fragmentation of the fibers of cigarette filters is visible when they are collected from the environment soon after being smoked. Smoked CBs are more damaged than unsmoked. This could make them more prone to detach microfibers to the environment during the degradation process.

CBs have a high degree of acetate substitution (2.6 to 2.8) which makes cellulose challenging to biological decomposition (Belzagui et al. 2021). Cellulose acetate is more susceptible to photodegradation than to biodegradation (Novotny et al. 2009). To be a source of food for organisms, cellulose acetate should be de-acetylated by chemical hydrolysis to a lower degree of substitution (~1).

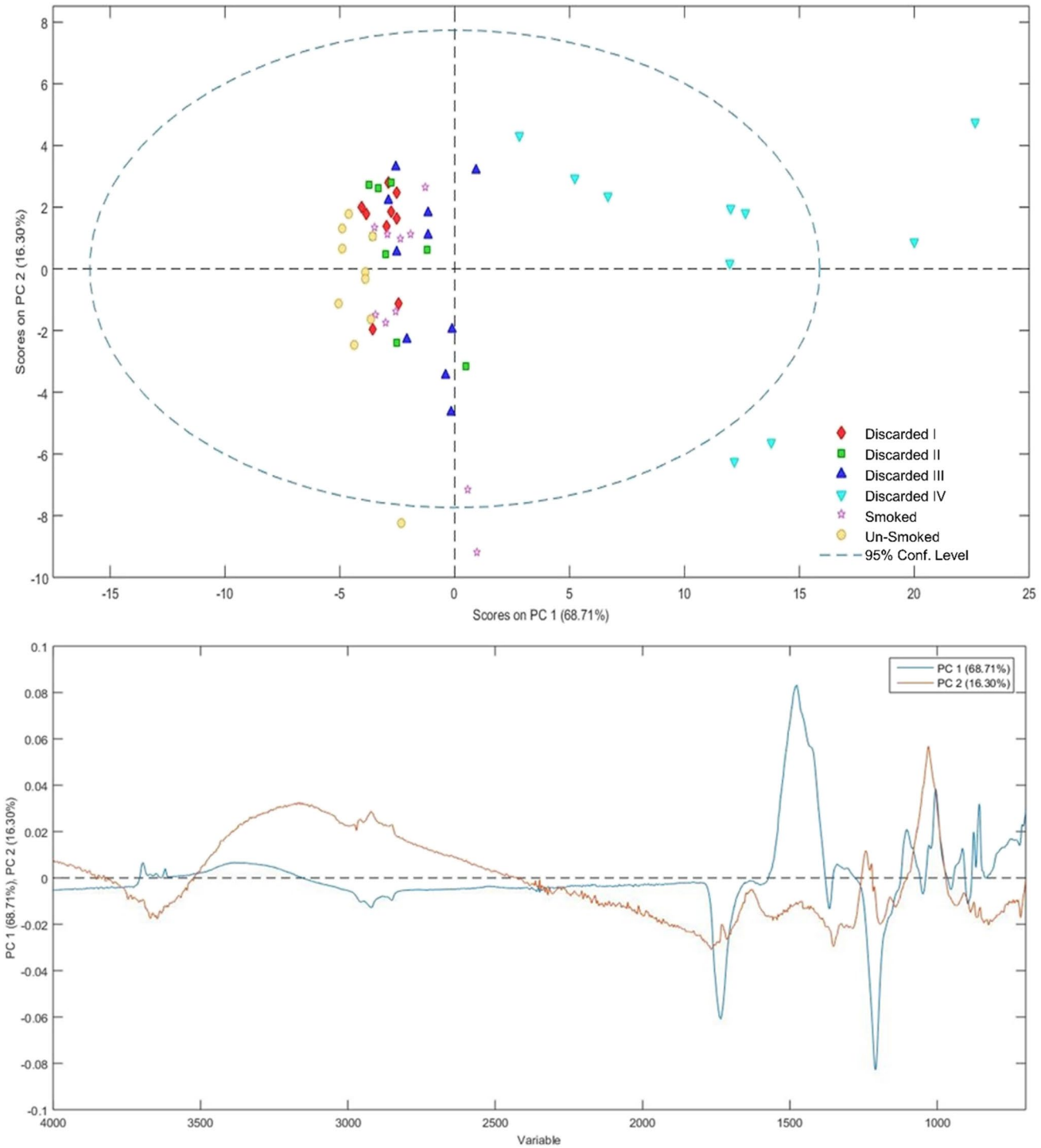


Fig. 5 Principal component analysis (PCA) of the six spectra processed with SNV and cantered at the average ($N = 10$ readings per degradation level I to IV, un-smoked, and smoked). Out of sixty readings, three outliers were removed. PC1 explains the variability of the most degraded CBs (IV) in relation to the other types. The largest loadings in PC1 point toward the crystalline region (1500–1300 cm^{-1}) as the most variable, suggesting this is the main difference between this degradation level and the other types. Lower pane: loadings

In addition, another factor that hinders biodegradation is the low nutrient content of sandy environments, especially nitrogen (N) and phosphorus (P) (Bonanomi et al 2015). Only recently, it was demonstrated that fungi can use this residue as a source of energy over long periods (Bonanomi et al 2020).

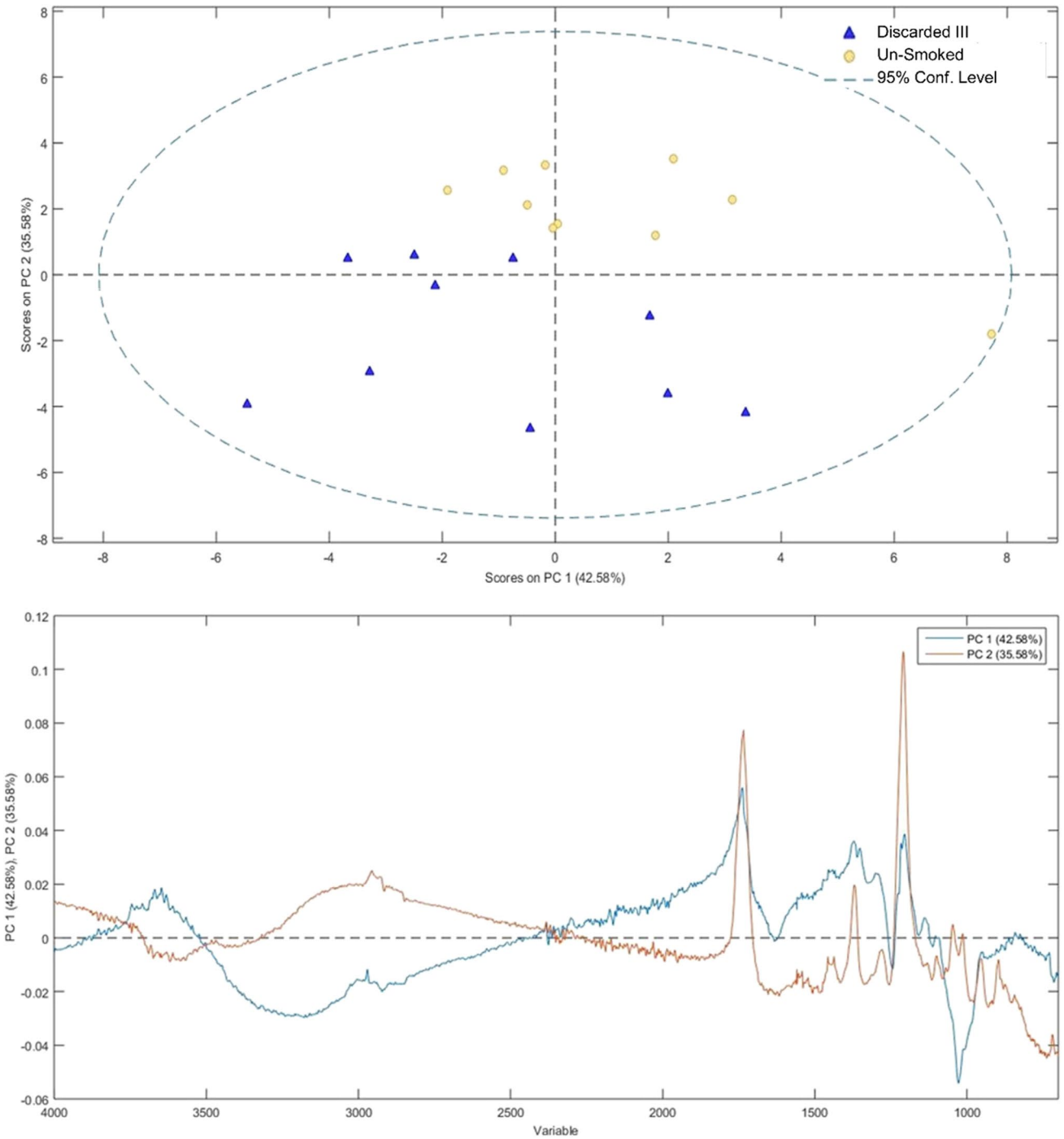


Fig. 6 Principal component analysis (PCA). Spectra of un-smoked and degradation level III, processed with SNV and cantered at the average ($N = 10$ readings of degradation level III and un-smoked). The largest loadings are at the ester characteristic region (C = O ~ 1700 e C-O ~ 1200), being the main responsible for the variations in PC2. Un-smoked samples remain at the positive side of PC2; therefore, suggesting the main difference between these two types of CBs is in the acetate group. This is a strong indication that degradation level III is already undergoing some type of environmental degradation. Lower pane: loadings

Haynes et al. (1991) studied the photodegradation of cigarette filters. The filters were placed in wired mesh cages a few inches above a flat roof in Tennessee, USA, where they were exposed to sun, wind, and rain. There was no soil contact or standing water. Although biodegradation was minimized, a mass loss between 10 and 20% was recorded after the first year, and 20 to 40% mass loss was measured after two years. The molecular weight determination of

exposed CBs fiber samples indicated that the polymer was partly de-acetylated, and its molecular weight decreased during the photodegradation process.

Environmental conditions have a strong impact on the degradation rate of materials (Puls et al. 2011). Cellulose molecules are known to be readily metabolized by microbial enzymes; however, plastic has great resistance to microbial decomposition (Zugenmaier 2004). Joly and Coulis (2018) compared the decomposition of plastic filters (cellulose acetate) against cellulose-based filters and found that filter decomposition varied depends on the type of filters, with cellulose filters decomposing faster than plastic ones. Cellulose acetate filters took between 7.5 and 14 years to break down completely, and this process widely depending on the disposal environments.

The processes undergone by CBs in beach environments at medium and long terms are only starting to be unveiled. It is possible that degradation of the paper and tobacco is much faster than the degradation of the cellulose acetate filter that accumulates in the sand when incorrectly disposed and not removed. The degradation of this part of the CBs is then much slower and can have different cycles according to the chemical decay phase the cellulose acetate is. Therefore, this process might be neither linear nor complete within a reasonable time span, allowing the CBs to remain in the environment and interact with toxic chemicals and biota for long periods.

Finally, we suggest that CBs are not recycled. The common and sole objective of all stakeholders should be to eliminate this type of residue from society and the environment. First, by correct discarding and as soon as possible, by zeroing generation. Recycling should not be a waste management option; the recuperation of cellulose acetate although technically viable and the object of various research projects (Moroz et al. 2021; Yousefi et al. 2021) poses a challenge to the achievement of a tobacco-free society.

Conclusion

Four levels of degradation were determined for cigarette butts collected on the beach. Cigarette butts exposed to beach sand environments lost length and mass along time. Physical fragmentation could be observed in SEM images when comparing un-smoked and smoked filters. FTIR-ATR of degraded cigarette butts indicated modifications in the spectra when compared to un-smoked cigarettes. Degradation is probably dependant on the break-down of specific cellulose acetate bonds.

CBs discarded in beach areas undergo physical changes as a result of the length of stay and the consequent weathering to which they are subjected. This fact was visually detected and allowed the suggestion of a degradation scale worth including and perfectioning in further studies. One of the possible uses of our scale (although it still needs further calibration) is for use in works that do not necessarily involve chemical analysis in the lab, or for situations in which people do not have the objective or means to set up experiments, but still wish to estimate different "ages" of CBs on their beach surveys. This could be a very useful information for activities related to beach clean-ups, environmental education, and other projects aiming at long term and/or wide areas surveys.

However, despite having an evident physical degradation, from a chemical point of view, this could only be detected at the most advanced level of degradation of our scale. This characteristic would allow evidencing the permanence of the polluting potential for long periods. CBs accumulate over time, approaching level IV and becoming more difficult to detect with the naked eye and to be removed from the environment for any of the known means of beach sand cleaning. During this time, CBs might go from degraded to disintegrated and generate an enormous amount of synthetic fibers that will remain available in the environment for an unknown period during which it will interact chemical and biologically.

The level IV of our visual scale was estimated to be available in beach sand for a period of five years. This level also presented the widest variation in length and mass within the level when compared to the other three. It would be interesting to work on criteria (physical, chemical, and toxicological) that allows for the subdivision of this level and try to find out what happens to CB after degradation level IV.

There is a clear disparity between the current knowledge about the presence of CBs (especially related to the amounts and main sources) in the most diverse locations, whether coastal or not, and the effect of these residues on the environment and its biota; despite the common consensus of the harmful effects that smoking causes to human health.

Environmental damage caused by any compound present in natural environments is often related to several factors, such as the ability to undergo leaching, dissolution, and the ability to chemically interact with other compounds. However, the length of stay in the environment is an extremely relevant factor, since the longer the length of stay, the greater the damage. When it is considered that CBs are not biodegradable and have high resistance to physical degradation processes, it is possible to infer the magnitude of the impacts they can cause, due to the slow and permanent release of the hundreds of toxic compounds they have.

Therefore, future projects on beaches and other environments could approach questions such as the number of CBs found in the different degradation level and impact/accumulation over time and how long do CBs stay in the sediment and leach pollutants to the environment.

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Author contribution Maria Christina B. Araújo, Monica F. Costa, and Jacqueline S. Silva-Cavalcanti were responsible for conceptualization, investigation, sampling, methodology, writing—original draft, writing-reviewing and editing. Armando C. Duarte, Vanessa Reis, Teresa A. Rocha-Santos, João Pinto da Costa, and Violeta Girão participated in development of methodology, formal analysis, investigation, and data validation. All authors read and approved the final manuscript.

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Data availability Not applicable.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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